

Introduction to Data Visualization

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(with help from Christina Maimone, Frank Elavsky, Steve Franconeri
and Adam Miller)





Edward Tufte's “Data to Ink Ratio”

The golden rule of visualization

Remove
to improve
(the **data-ink** ratio)



Initial Considerations

First, think carefully about these:

1. Choose exploratory vs. explanatory visualization.
2. Use storytelling to share your message.
3. Use your tools wisely (salience, symbols, colors, dimensions, axes/grids).
4. Carry these important suggestions in your back pocket.

Improve Data-to-Ink Ratio

Increase Efficiency

Ensure Visual Quality

Consider Accessibility

Reduce Clutter

Organize and Guide

Two great resources: [Nature Methods points of view](#), [Google material design principles](#)



Important Suggestions

Improve Data-to-Ink Ratio

Is this [shape] necessary?

Ensure Visual Quality

Is this geometry telling the truth?

Reduce Clutter

Is this color choice or layout necessary?

Increase Efficiency

Is it too hard or time consuming to read?

Consider Accessibility

Is this colorblind safe? Is the font size large enough?

Organize and Guide

Should I regroup my data? Can I add helpful text?

Exploration

Help YOU learn about your data

Audience

You and your collaborators

**Number of
Visualizations**

Many

**Visualization
Message**

Unknown

Formatting

Not important

Exploration

Help YOU learn about your data

Audience

You and your collaborators

**Number of
Visualizations**

Many

**Visualization
Message**

Unknown

Formatting

Not important

Explanation

Help OTHERS learn about your data

Pick one

Probably 1 per dataset

Why include the visualization?

Important, possibly restricted



Storytelling

Communication: (noun) a process by which info is exchanged between individuals through a common system of symbols, signs, behavior.

Humans communicate via **stories.**



Storytelling

a

13 53 81 29 25

22 68 62 24 78



Storytelling

a

13 53 81 29 25

22 68 62 24 78

b

• • ● • •

• ● ● • ●

• 0–30 • 31–60 ● 61–100

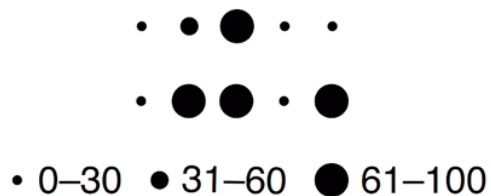


Storytelling

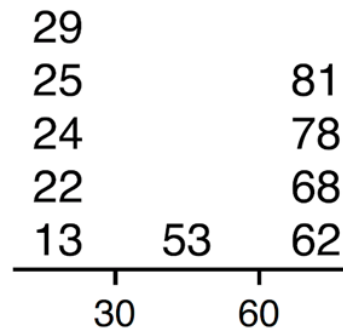
a

13 53 81 29 25
22 68 62 24 78

b



c



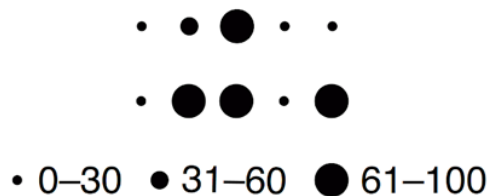


Storytelling

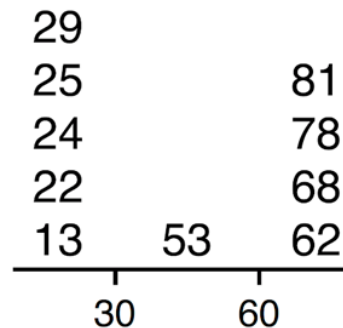
a

13 53 81 29 25
22 68 62 24 78

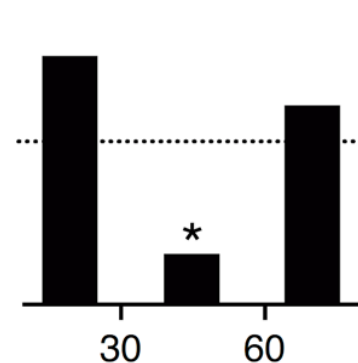
b



c

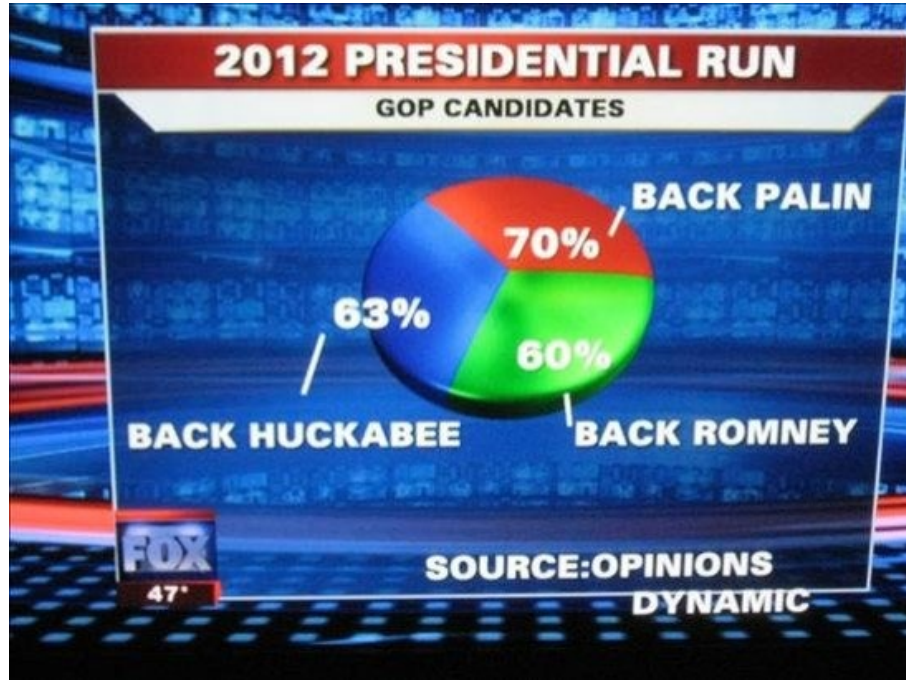


d



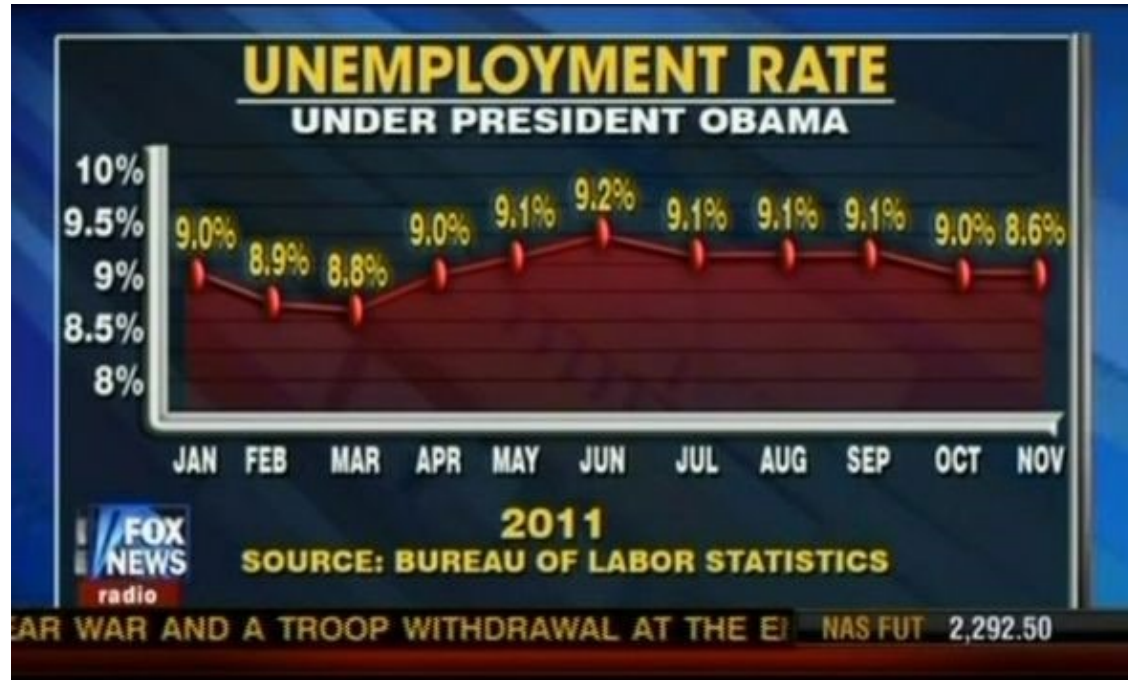


Use Your Tools Wisely!





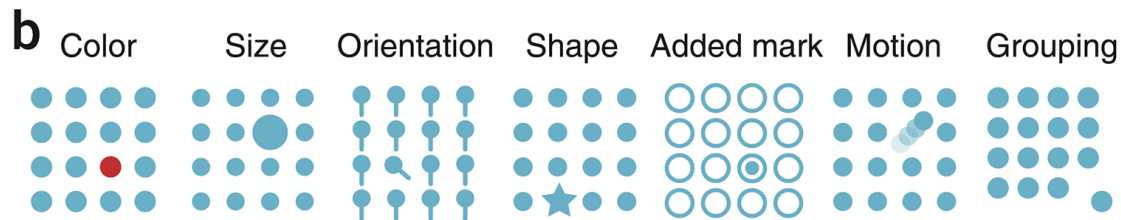
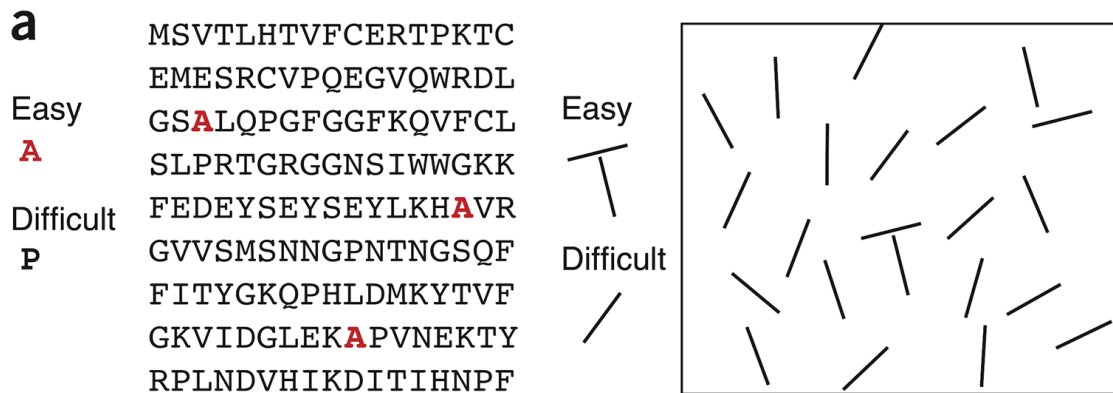
Use Your Tools Wisely!





Your Tools: Saliency

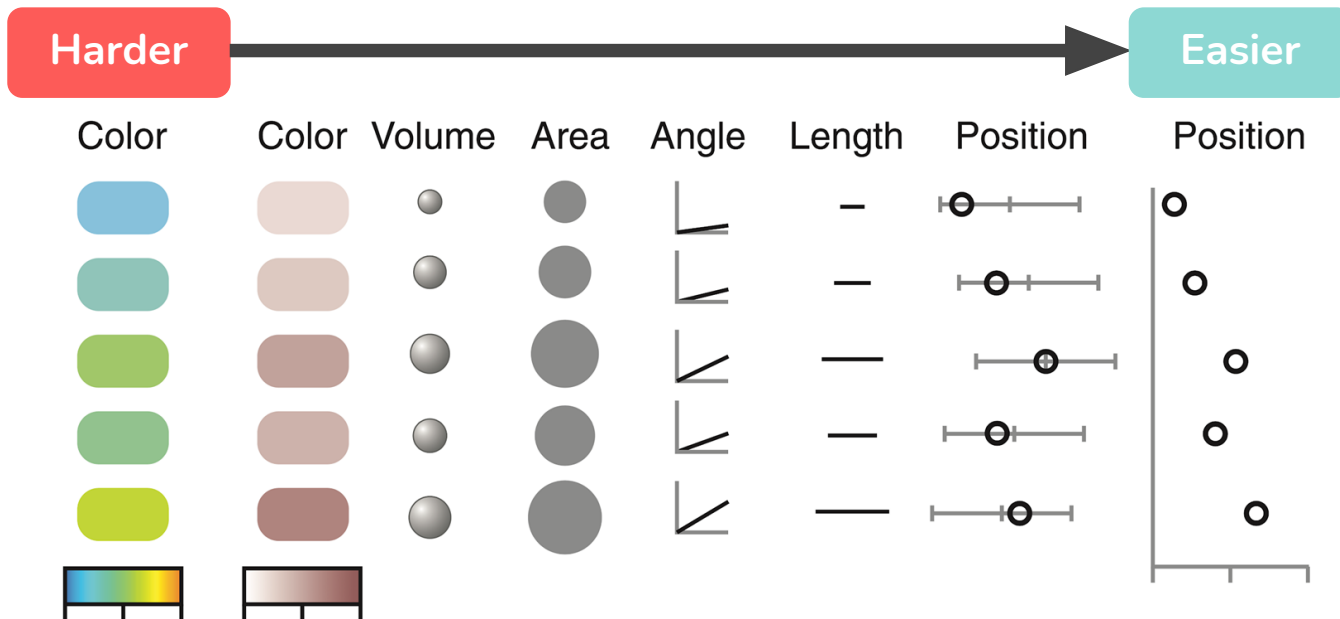
Guide the viewer to your result.





Your Tools: Saliency

Use easy-to-estimate visual representations





Your Tools: Salience

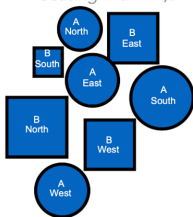
Organize for comparison. Consider time to digest.

Harder

Easier

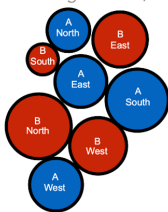
Shape

Focusing within A,B



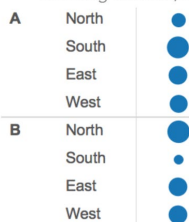
Hue

Focusing within A,B



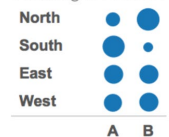
Position V or H

Focusing within A,B



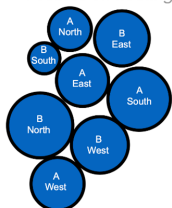
Position V+H

Focusing on Both A+B!



Just labels

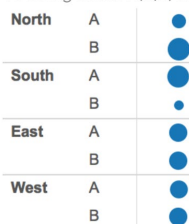
Focuses on nothing



Focusing within N,S,E,W

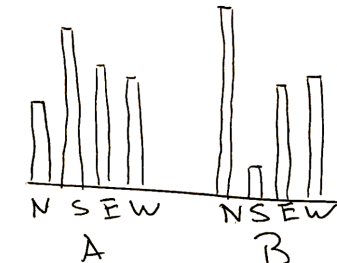
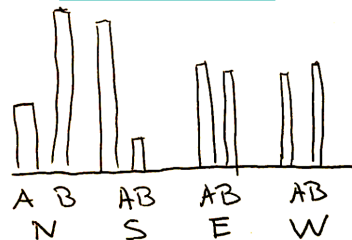
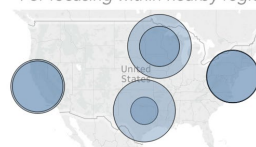


Focusing within N,S,E,W



Maps are Position V+H

For focusing within nearby regions

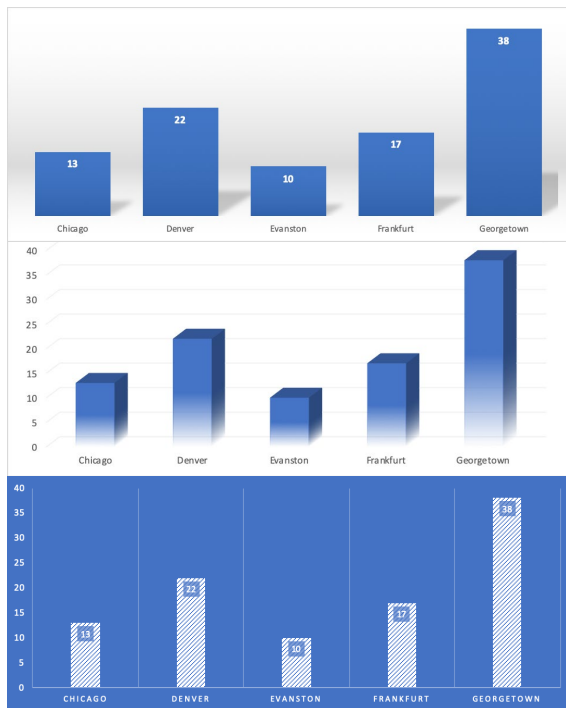


Slide adapted from Steve Franconeri (Northwestern)



Your Tools: Salience

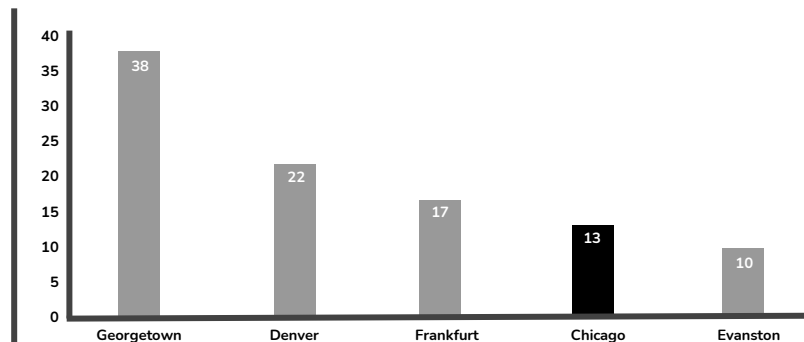
Keep it simple.



No shadows

No gradients

No patterns



Yes

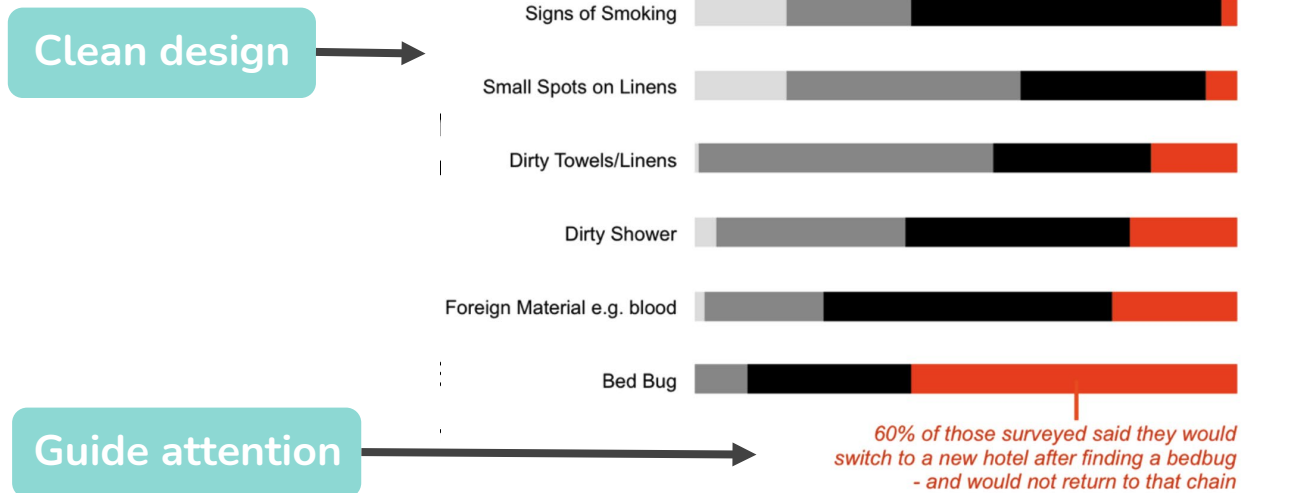
Expert tips: Order bars by value (if categories aren't ordered), and guide the reader to important value(s)



Your Tools: Salience

Use descriptive labels (and titles).

Bedbugs are the #1 cause of **lost guests**



Slide adapted from Steve Franconeri (Northwestern)



Your Tools: Salience

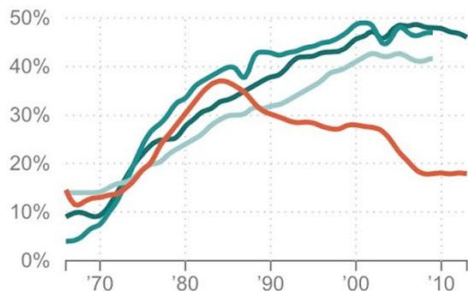
Try direct labels instead of legends.



What Happened To Women In Computer Science?

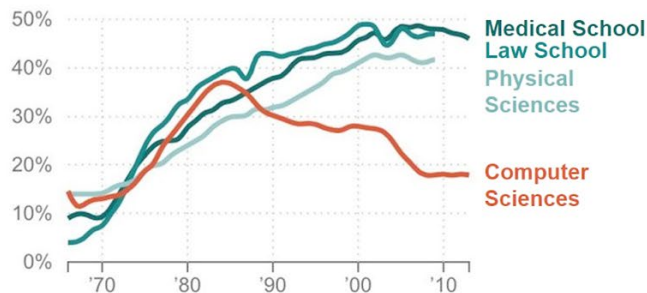
% Of Women Majors, By Field

Medical School Law School
Physical Sciences Computer science



What Happened To Women In Computer Science?

% Of Women Majors, By Field

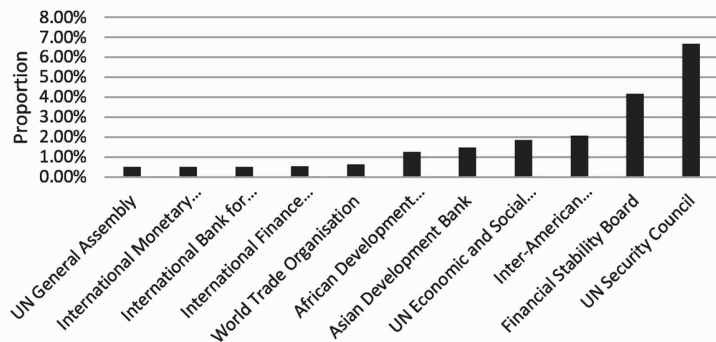




Your Tools: Salience

Make sure labels are legible (orientation, font size, etc.).

Proportion of members of developing countries
in international organizations



No

Developing countries make up less than 7% of international organization membership.

Proportion of members of developing countries in international organizations, most current year of data (2015, 2016)

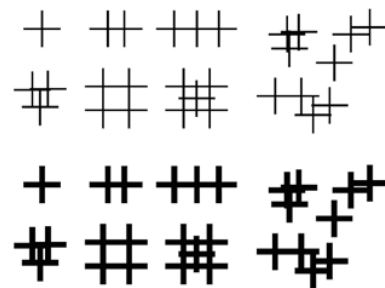
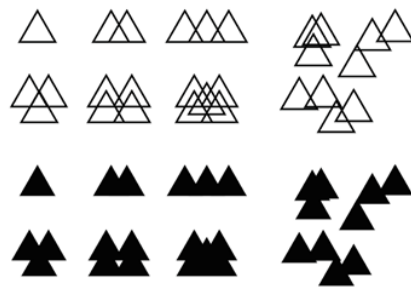
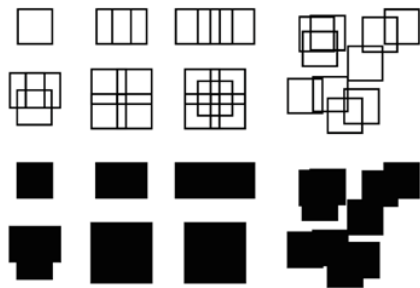
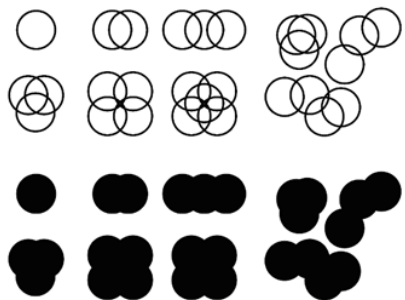


Yes



Your Tools: Symbols

Open circles are the most flexible.

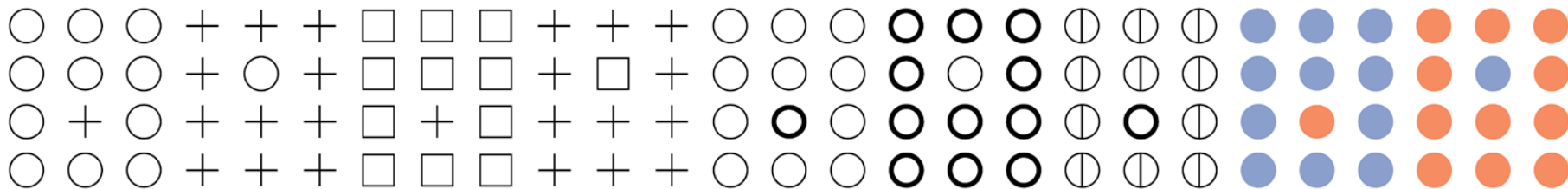




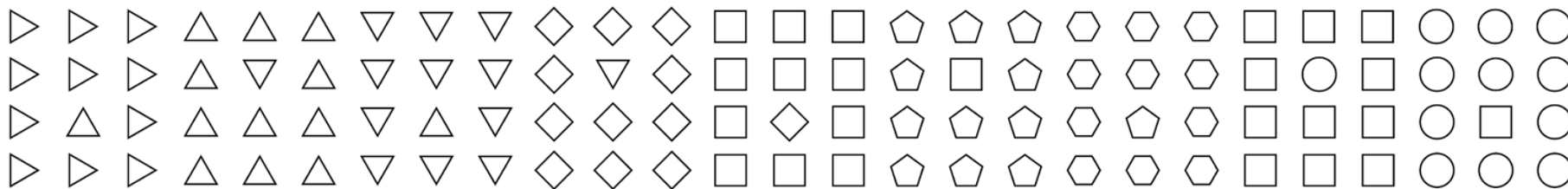
Your Tools: Symbols

Form strong visual boundaries.

Strong visual boundaries



Weak visual boundaries

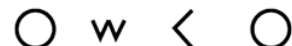




Your Tools: Symbols

Form strong visual boundaries.

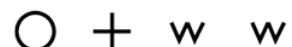
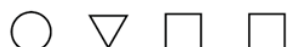
A N T A



A T G T



A C N N



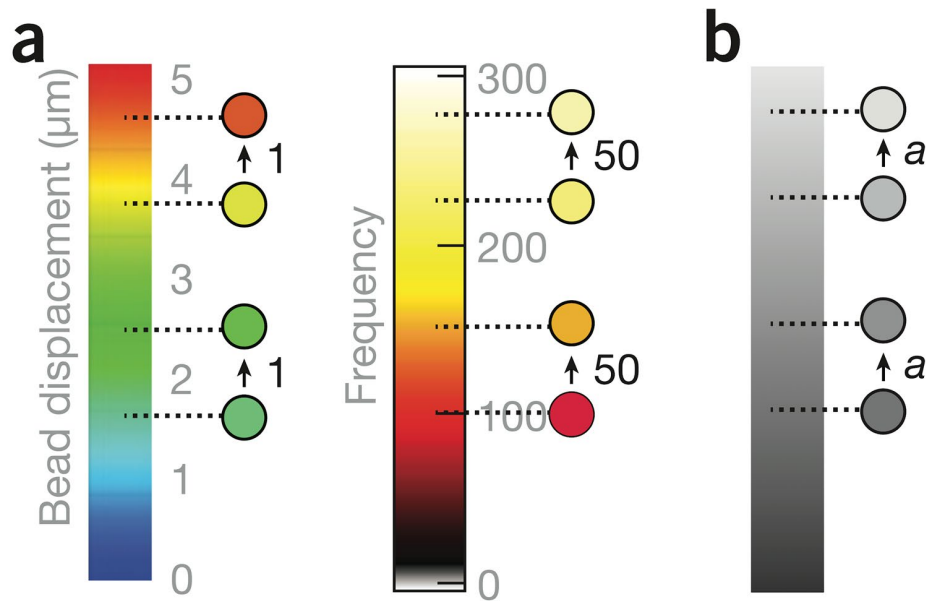
A A A A





Your Tools: Colors

Choose colormaps wisely, but note that color is not ideal for representing quantitative data.

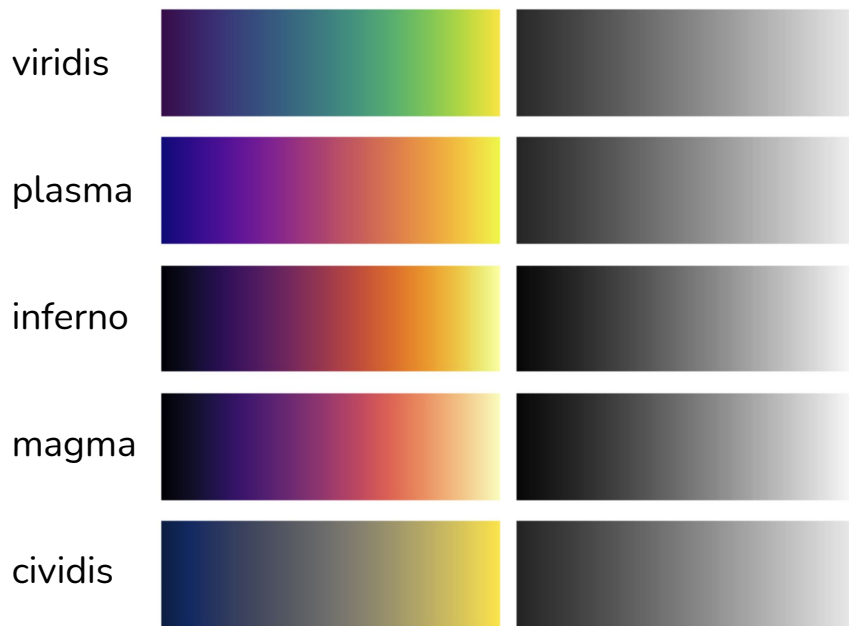




Your Tools: Colors

What does it look like in greyscale? Is it colorblind safe?

Perceptually uniform sequential colormaps



<https://www.color-blindness.com/coblis-color-blindness-simulator/>
<https://colorbrewer2.org/>
<https://coolors.co/>

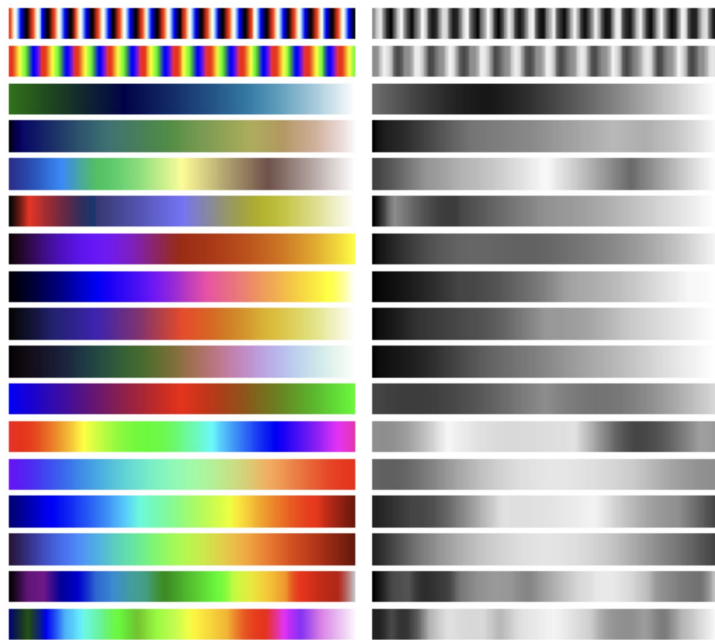
[matplotlib](https://matplotlib.org/)



Your Tools: Colors

What does it look like in greyscale? Is it colorblind safe?

There are LOTS of (often very bad) colormaps out there. Be careful!





Your Tools: Colors

Colors can have meaning.

Good

Bad

Banana

Sky



Your Tools: Colors

Colors can have meaning.

Good

Bad

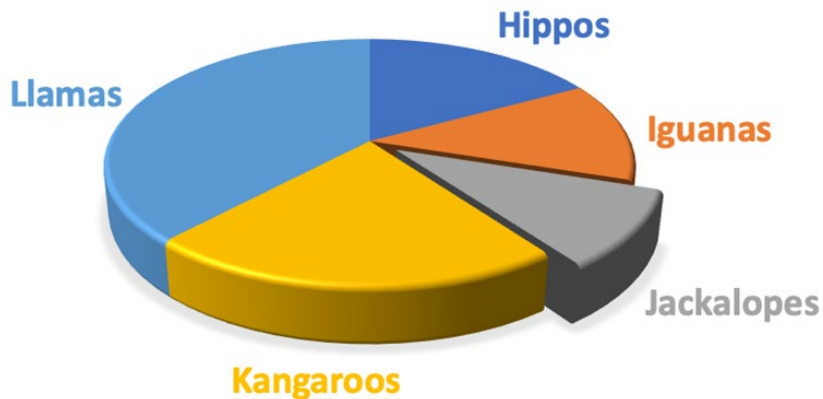
Banana

Sky

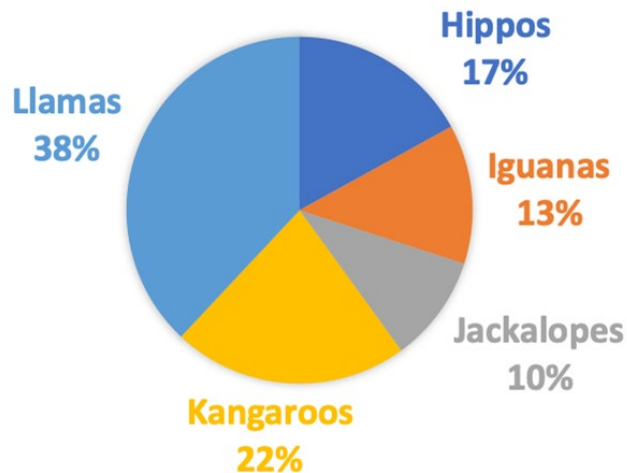


Your Tools: Dimensions

Stick to 2D whenever possible.



No



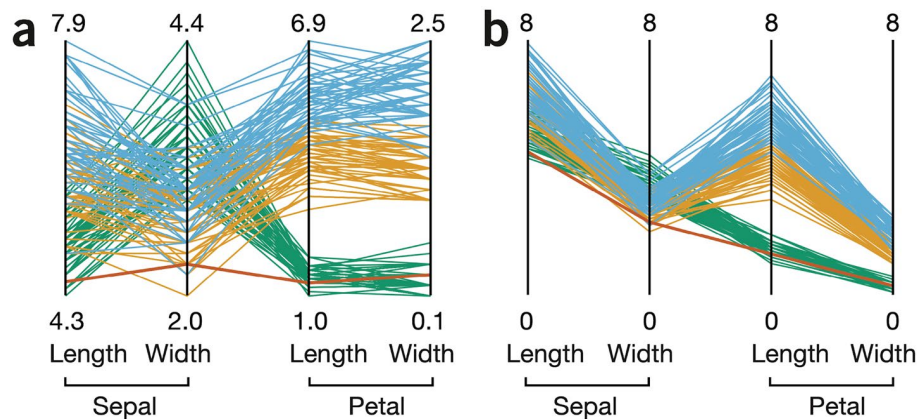
Yes



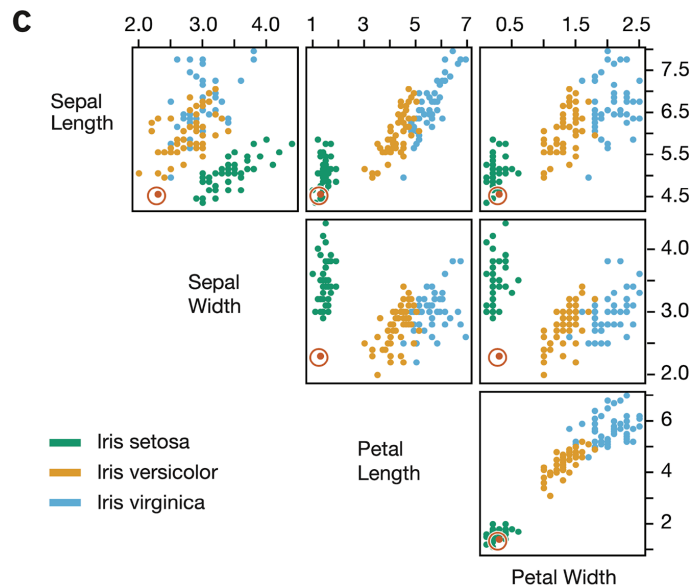
Your Tools: Dimensions

Combine multiple dimensions in 2D.

Parallel coordinate plots



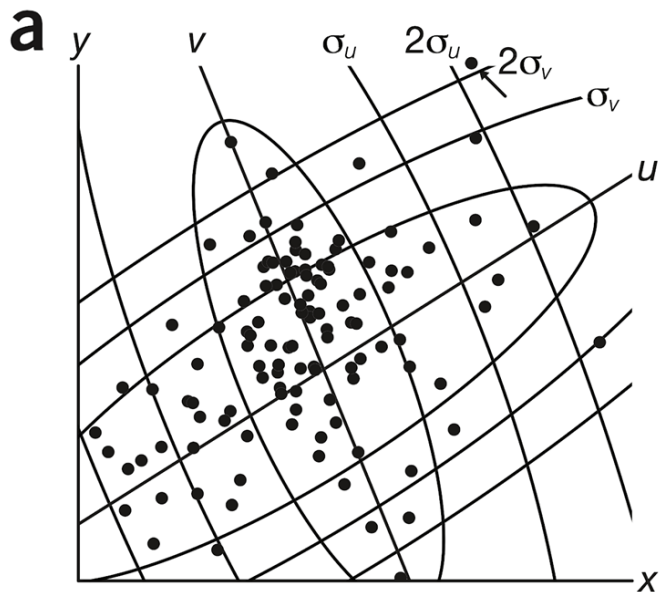
Corner/matrix plot





Your Tools: Axes & Grids

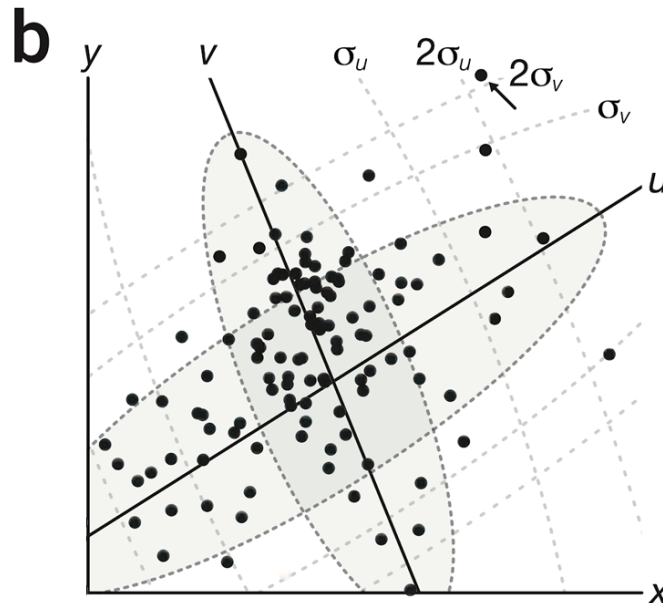
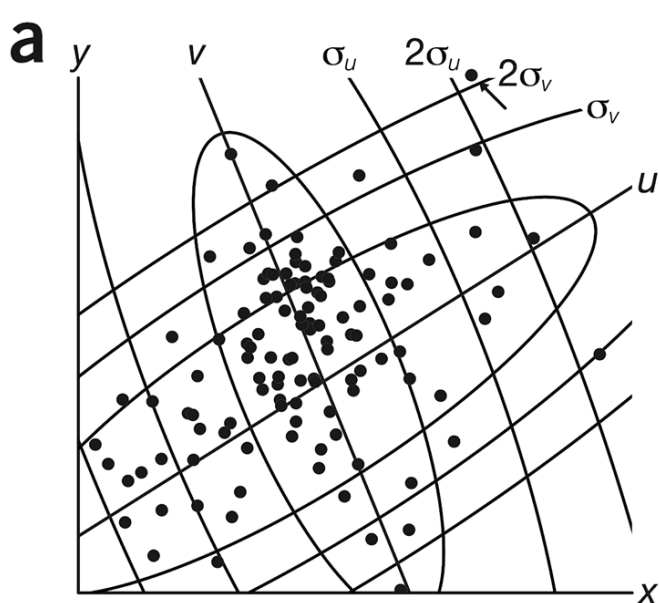
Make navigational elements visually distinct.





Your Tools: Axes & Grids

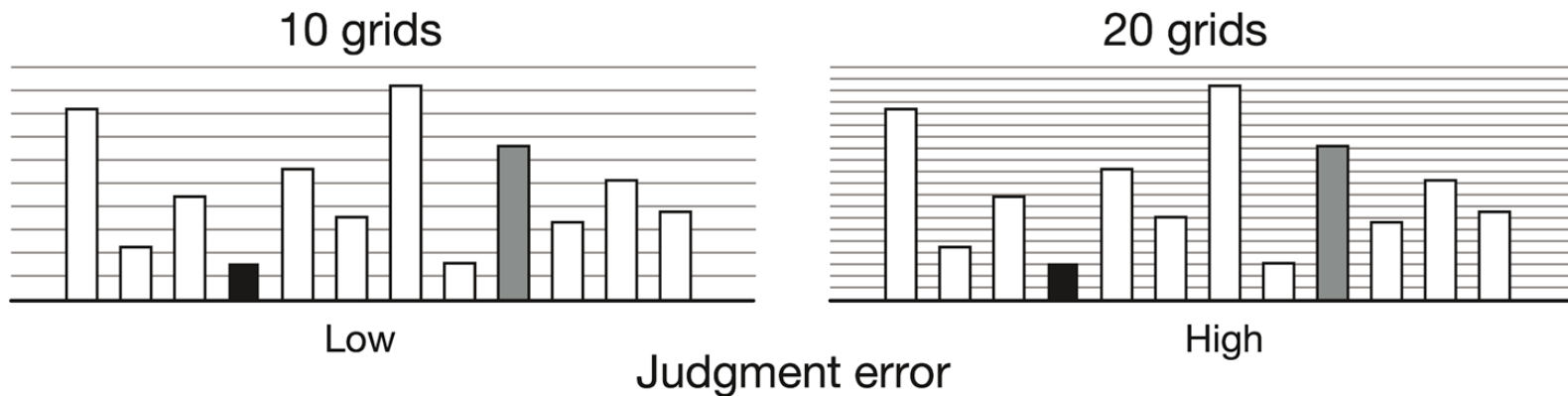
Make navigational elements visually distinct.





Your Tools: Axes & Grids

Use grid lines judiciously.





Summary

First, think carefully about these:

1. Choose exploratory vs. explanatory visualization.
2. Use storytelling to share your message.
3. Use your tools wisely (salience, symbols, colors, dimensions, axes/grids).
4. Carry these important suggestions in your back pocket.

Improve Data-to-Ink Ratio

Increase Efficiency

Ensure Visual Quality

Consider Accessibility

Reduce Clutter

Organize and Guide

Two great resources: [Nature Methods points of view](#), [Google material design principles](#)



Questions?

(I have a hands-on Python walkthrough next.)



Hands on with Python

IntroToDataVis.ipynb

File Edit View Insert Runtime Tools Help All changes saved

+ Code + Text

RAM Disk

Editing

Hands-on with Python + matplotlib

What is wrong with this figure?

Poor Usage of Characteristics in Graphs

Let's agree that this is a monstrosity. Now, how do we improve it?

```
[1] import numpy as np
import pandas as pd
import matplotlib
from matplotlib import pyplot as plt

%matplotlib inline
```

1. Read in the data

I'm using pandas.

```
[4] url = 'https://raw.githubusercontent.com/ageller/IDEAS_FSS-Vis/master/matplotlib/bar/bar.csv'
data = pd.read_csv(url)
data
```

	Label	Value
0	Encoding Visual Meaning	10.0
1	Truncating Axes	5.0
2	Data to Ink Ratio	0.5
3	Efficiency of Copation	0.0

[Click here to view notebook.](#)